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A Multihead Attention Deep Learning Algorithm to Detect Amblyopia Using Fixation Eye Movements

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Objective: To develop an attention-based deep learning (DL) model based on eye movements acquired during a simple visual fixation task to detect amblyopic subjects across different types and severity from controls.

Design: An observational study.

Subjects: We recruited 40 controls and 95 amblyopic subjects (anisometropic = 32; strabismic = 29; and mixed = 34) at the Cleveland Clinic from 2020 to 2024.

Methods: Binocular horizontal and vertical eye positions were recorded using infrared video-oculography during binocular and monocular viewing. Amblyopic subjects were classified as those without nystagmus (n = 42) and those with nystagmus with fusion maldevelopment nystagmus (FMN) or nystagmus that did not meet the criteria of FMN or infantile nystagmus syndrome (n = 53). A multihead attention-based transformer encoder model was trained and cross-validated on deblinked and denoised eye position data acquired during fixation.

Main Outcome Measures: Detection of amblyopia across types (anisometropia, strabismus, or mixed) and severity (treated, mild, moderate, or severe) and subjects with and without nystagmus was evaluated with area under the receiver-operator characteristic curves, area under the precision–recall curve (AUPRC), and accuracy.

Results: Area under the receiver-operator characteristic curves for classification of subjects per type were 0.70 ± 0.16 for anisometropia (AUPRC: 0.72 ± 0.08), 0.78 ± 0.15 for strabismus (AUPRC: 0.81 ± 0.16), and 0.80 ± 0.13 for mixed (AUPRC: 0.82 ± 0.15). Area under the receiver-operator characteristic curves for classification of amblyopia subjects per severity were 0.77 ± 0.12 for treated/mild (AUPRC: 0.76 ± 0.18), and 0.78 ± 0.09 for moderate/severe (AUPRC: 0.79 ± 0.16). Th area under the receiver-operator characteristic curve for classification of subjects with nystagmus was 0.83 ± 0.11 (AUPRC: 0.81 ± 0.18), and the area under the receiver-operator characteristic curve for those without nystagmus was 0.75 ± 0.15 (AUPRC: 0.76 ± 0.09).

Conclusions: The multihead transformer DL model classified amblyopia subjects regardless of the type, severity, and presence of nystagmus. The model's ability to identify amblyopia using eye movements alone demonstrates the feasibility of using eye-tracking data in clinical settings to perform objective classifications and complement traditional amblyopia evaluations.

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Vision plays a vital role in a child's development. Amblyopia poses significant risks as the primary cause of preventable visual impairment in 1 eye, affecting around 3% to 4% of children.[1–5] Amblyopia arises from unequal visual input early in life due to conditions like strabismus, anisometropia, or cataract, leading to reduced visual acuity, contrast sensitivity, stereopsis deficits, and eye movement abnormalities such as unsteady fixation.[6] Amblyopia can impact daily tasks,[7–15] academic performance, and psychological well-being, with an annual economic burden of around US $7.4 billion.[16] Early intervention is essential as the visual system's plasticity decreases with age, yet identifying amblyopia in children, especially those without obvious signs like strabismus, remains challenging.[17–19] Thus, tremendous resources are allocated for vision screening and timely amblyopia detection.[20,21]

Clinical diagnosis of amblyopia relies on detecting reduced visual acuity in 1 eye after correcting amblyogenic factors. Visual acuity tests are commonly used in ophthalmology offices and vision screenings by pediatricians and school nurses. Instrument-based devices, like photoscreeners, detect amblyogenic risk factors, although they often have high false-positive rates and low specificity.[22–24] Recent studies have shown that the cover test, which detects strabismus, or visual acuity testing along with instrument-based screening increases the specificity for amblyopia detection.[25,26] However, visual acuity assessments are subjective, and the cover test relies on real-time behavioral

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observations, both requiring examiner-dependent skills without providing objective data for offline analysis.[27–29]

Eye movement data obtained during a simple visual fixation task can serve as an objective tool to aid in identifying amblyopia.[30] Fixation eye movements (FEMs), which are typically imperceptible, include small involuntary motions such as microsaccades, drifts, and ocular tremor.[31,32] In amblyopia, these physiological FEMs become unstable,[33–37] manifesting as increased amplitude of fixational saccades, drifts, and nystagmus (Fig 1). Studies have demonstrated a strong correlation between FEM abnormalities and visual sensory function deficits, including visual acuity, contrast sensitivity, suppression, and stereoacuity in amblyopia.[30,33,36–45] Our previous research revealed that fixation instability differs between the amblyopic and fellow eyes and varies under different viewing conditions.[30,37,46] Specifically, during binocular viewing (BV) and monocular viewing (MV), amblyopic eyes show greater fixation instability compared to fellow eyes and controls. Previous studies have applied classical machine learning models such as random forests in analyzing eye movements in various contexts, such as reading, viewing naturalistic scenes, and watching movies.[47–49] To the best of our knowledge, this is the first study to develop and validate a clinical informatics tool to identify amblyopia using FEM data together with a new attention-based deep learning (DL) model (Fig 2). Further, the results from this study validate the generalizability of this DL model to classify different subpopulations of amblyopic subjects based on their severity, type, and nystagmus status with relatively high accuracy.

Methods

Ethics and Institutional Governance Approvals

This study was reviewed and approved by the Cleveland Clinic Institutional Review Board. Written informed consent was obtained from each participant or parent/legal guardian for this study as mandated by the Declaration of Helsinki.

Study Cohort and Demographics

Amblyopia is defined as subnormal-corrected distance visual acuity in the absence of any structural optic nerve, retinal, or visual pathway abnormalities. All subjects categorized within the amblyopic cohort exhibited at least a 3-line difference in visual acuity between the 2 eyes at the time of diagnosis. The inclusion criteria for amblyopic subjects were specified as the presence of amblyopia attributable to anisometropia, strabismus, or a combination of anisometropia and strabismus (labeled as "mixed mechanism amblyopia" in the study). The exclusion criteria included any coexisting ocular or systemic disease, congenital infections or malformations, developmental delay, or bilateral amblyopia. Subjects with idiopathic infantile (congenital motor or sensory anomaly) nystagmus were also excluded. Idiopathic infantile nystagmus is characterized by an increasing-velocity slow-phase component of the nystagmus, or alternatively, pendular nystagmus (no fast-phase), which can affect visual acuity measurements.[50] We also excluded subjects with deprivation amblyopia, as it is relatively rare compared to anisometropic and strabismic amblyopia. Including deprivation amblyopia could introduce variability due to its distinct causes and presentation, potentially skewing results, given the limited sample size. Excluding it ensures a more homogeneous dataset, enhancing the accuracy and generalizability of the artificial intelligence algorithms for detecting the more common forms of amblyopia. Control subjects in the study were chosen based on the absence of any ocular or systemic abnormalities affecting visual acuity, except for refractive errors. Participants' ages ranged from 3 to 72 years. Subjects' demographic data, including age, sex, and race, were collected based on self-report.

All subjects recruited in the study had a comprehensive eye examination. The visual acuity of the fellow and amblyopic eye and stereoacuity, cycloplegic refraction, and strabismus angle measurements were extracted from a retrospective chart review. Visual acuity was measured monocularly, starting from the right eye, using the participant's optimal spectacle correction with Snellen linear optotype. For children <6 years old, crowding bars HOTV optotypes were preferred and used over picture optotypes (Allen optotypes with crowding bars presented with the commercially available computer-based system Accomodata Stimuli). Visual acuity was assessed at a distance of 20 feet, and the value was recorded only if the patient could correctly identify all the letters or symbols on the line. Stereoacuity was measured with the Titmus Stereoacuity Test (log arcsec). Patients with no stereoacuity were assigned a value of 3.85 log arcsec.[34]

Categorization of Amblyopic Subjects

Type of Amblyopia. The clinical categorization of amblyopia subtype (strabismic, anisometropic, and mixed) and severity (mild, moderate, and severe) was based on PEDIG studies.[51] Strabismic amblyopia was diagnosed if there was heterotropia at distance and near fixation (with or without spectacles), a history of strabismus surgery, or previous strabismus that resolved with glasses and surgery. Anisometropic amblyopia was diagnosed if there was a significant difference in spherical equivalent (≥0.50 diopters [D]) or astigmatism (≥1.50 D) between the 2 eyes. Mixed mechanism amblyopia was diagnosed if criteria for strabismus were met, along with a ≥1.00 D difference in spherical equivalent or ≥1.50 D difference in astigmatism between the eyes.

Severity of Amblyopia. The severity of amblyopia was determined based on the visual acuity deficit of the amblyopic eye at the time of eye movement recordings. Amblyopia was classified as treated if the worse eye visual acuity was ≤0.09 logMAR, mild if the visual acuity was between 0.09 and <0.30 logMAR, moderate if visual acuity fell between ≥0.30 and <0.70 logMAR, and severe if visual acuity was ≥0.70 logMAR per the PEDIG studies.[52–54]

Eye Movement Recording and Analysis

Eye movements were measured using the EyeLink1000 plus (SR Research), a noninvasive, high-resolution video-oculography tracker that has a spatial resolution of 0.01° and a temporal resolution of 500 Hz, as described in previous publications.[11,30,55] Subjects wore their corrective lenses, if applicable, for the experiments to achieve their best-corrected vision. Subjects were seated comfortably in a dark room, and the subject's head was stabilized on a chin rest 84 cm away from an liquid crystal display computer monitor where the visual stimuli were displayed. We have created stimuli featuring cartoon images and audio sounds to enhance engagement. Each trial began with a large, colorful animal cartoon along with animal sounds, and its image shrank in 3 rapid steps. A similar technique has been used for visual evoked potentials testing in young children.[56] Monocular calibration and validation of the right and left eyes were done using a 5-point

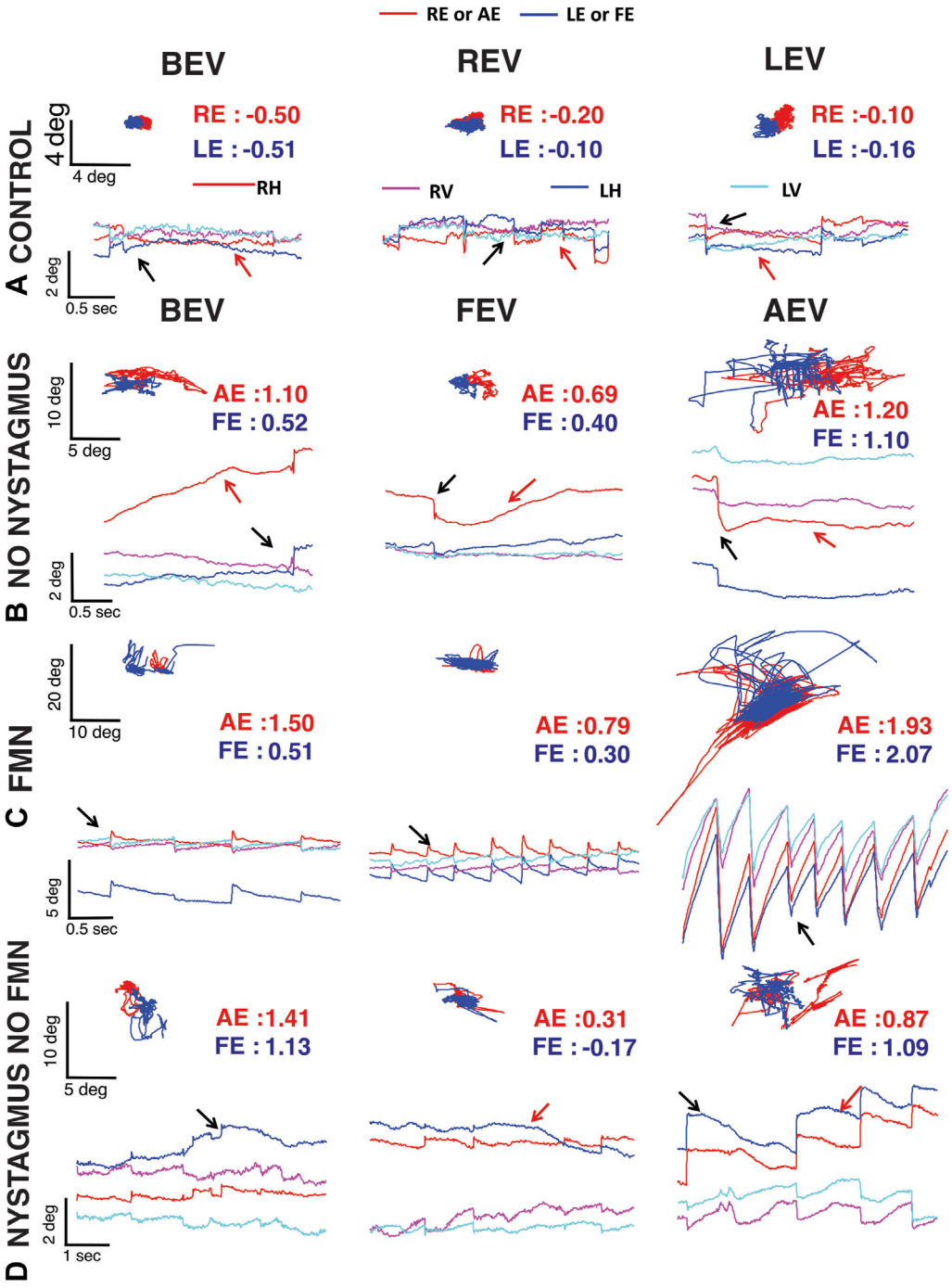


Figure 1. Fixation stability and time series of FEM traces (bottom) in a control **(A)** and amblyopia subjects without nystagmus **(B)**, with FMN **(C)**, and nystagmus without FMN **(D)** obtained during binocular viewing and monocular right and left eye viewing conditions. For amblyopic subjects, monocular viewing is labeled as fellow eye viewing and amblyopic viewing conditions. AE = amblyopic eye; AEV = amblyopic eye viewing; BV = binocular viewing; FE = fellow eye; FEV = fellow eye viewing; FMN = fusion maldevelopment nystagmus; LE = left eye; LEV = left eye viewing; LH = left horizontal eye position; LV = left vertical eye position; RE = right eye; REV = right eye viewing; RH = right horizontal eye position; RV = right vertical eye position.

constellation of horizontal and vertical targets per the manufacturer's guidelines.[30] After calibration, FEMs were recorded as the subject fixated a white circular target (0.5° visual angle) projected against a black background on the liquid crystal display monitor (30-inch diameter, resolution 2560 × 1600 pixels at 60 Hz, and brightness 350 cd/m^2). A total of 3 trials were recorded with each trial of 45 second durations. Each trial was recorded with binocular viewing (BV) or the left/right eye viewing with the other eye covered (labeled as MV) in a randomized order. The MV trials recorded the information about the viewing eye and nonviewing eye with fellow eye viewing (FEV) and amblyopic eye viewing (AEV) conditions. An

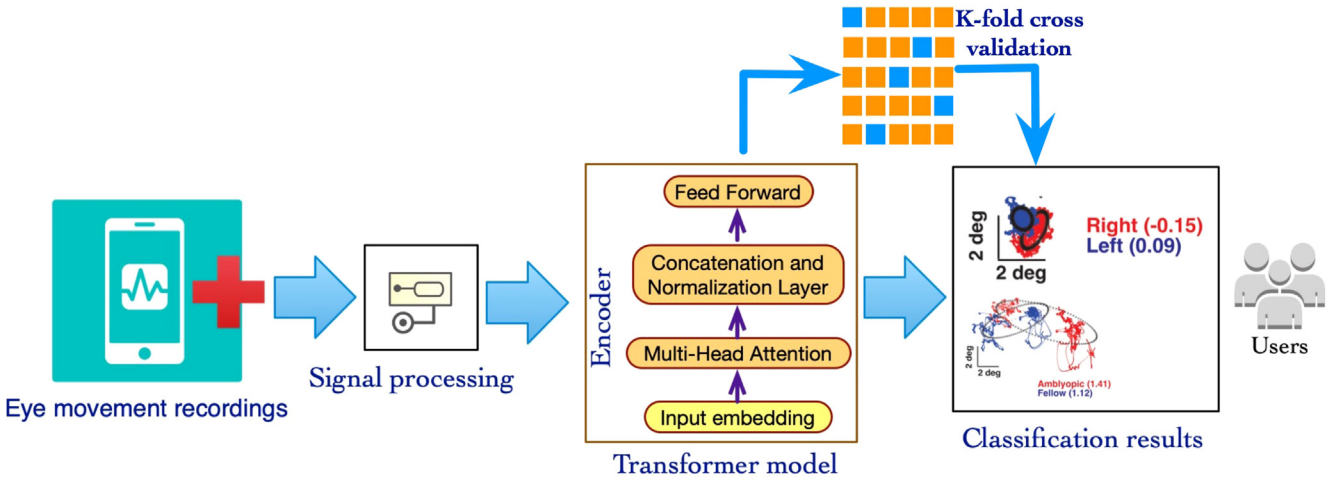


Figure 2. Flowchart depicting the steps in building and validating the model for classification of FEMs. FEMs = fixation eye movements.

infrared permissive filter was used to block visible light while allowing the nonviewing eye to be tracked. Our trials were designed to last 45 seconds to ensure the collection of high-quality data. The experimenter continuously monitored gaze positions in real-time during the sessions to ensure proper fixation was maintained. Additionally, stimuli presentation software was utilized to track gaze positions continuously throughout the experiment. An animal sound was employed to recapture the child's attention if their gaze deviated beyond a 4 × 4° fixation window for more than 500 milliseconds. If the subject failed to fixate on the target for >10 seconds within the 45-second trial, the trial was repeated. Approximately 10% of the study participants <5 years of age had repeat trials.

Fixation Eye Movement Data Analysis. We developed a customized graphical user interface script that allowed data visualization and manual verification that subjects were attending to the fixation target during data analysis. Blinks and noise were eliminated using the Engbert and Kleigl algorithm.[57] This automated algorithm was used to identify fixational saccades and quick phases of nystagmus.[57–59] Drifts and slow phases were defined as epochs between fixational saccades and quick phases in patients without and with nystagmus, respectively.[11,37]

Fixation stability was quantified using bivariate contour ellipse area, which encompasses a given proportion, usually 68%, of eye position samples during a fixation trial.[60] We and others have previously shown that the fixation instability in amblyopia arises from alterations in physiologic FEMs and nystagmus, defined as repetitive cycles of a fast-phase saccade and a slower, prolonged, slow-phase linear or decreasing-velocity drift of eye position during attempted steady fixation.[35,36,40,61] Bivariate contour ellipse area is a global measure that reflects the contributions of all FEM components. However, bivariate contour ellipse area analyses alone do not readily identify the presence of nystagmus and do not provide quantitative parameters of different FEM abnormalities.[37,62] Thus, the eye position traces obtained during BV, FEV, and AEV were evaluated to categorize the amblyopic subjects per their FEM waveforms (Fig 1). Subjects were classified based on the presence or absence of nystagmus into those without nystagmus (n = 42), those with nystagmus with fusion maldevelopment nystagmus (FMN) (n = 16), and those with nystagmus that did not meet the criteria of FMN (n = 37), for a total of 53 subjects.

Figure 1 plots the time series data of eye positions of the right eye and left eye acquired over a 5 second epoch during BV, FEV, and AEV. It also depicts the 2D Cartesian plot of horizontal and vertical eye positions of the right and left eyes acquired during fixation over a period of 45 seconds in control and 3 amblyopic subjects. In a control child (Fig 1A), the log 68% bivariate contour ellipse area values of the right and left eyes are lower, with less eye position scatter (stable fixation) than all amblyopic subjects (Fig 1B–D). In controls, the FEMs comprise microsaccades, which are binocular conjugate movements with an amplitude of <1° that occur at 1 to 2 Hz frequency (fast FEMs—black arrows) separated by intersaccadic drifts (slow FEMs—red arrows) (Fig 1A). Figure 1B depicts fixation stability and FEMs obtained from an amblyopic child without nystagmus. The amplitude of fixational saccades (black arrow) and intersaccadic drift velocity of the amblyopic eye (red arrow) are increased. Figure 1C depicts FEMs obtained from an amblyopic child with FMN. The hallmark of FMN is the reversal in the direction of the quick phase under MV, with the slow phases directed nasally with respect to the viewing eye. Note that the subject has manifest FMN, as nystagmus is also evident in BV. Figure 1D depicts FEMs obtained from an amblyopic child with nystagmus that does not meet the criteria of FMN. The nystagmus has decreasing/linear slow-phase velocities with dynamic overshoots of quick phases, differentiating it from those with congenital motor nystagmus with increasing slow-phase velocities.[39] The nystagmus is more conspicuous under MV particularly during AEV condition; however, there was no reversal in the direction of the nystagmus under MV. The nystagmus in this subject was labeled as nystagmus without FMN.

Power Calculations. The sample size was determined based on our previously published study examining fixation instability under MV and BV conditions in individuals with amblyopia and controls.[30] The estimated effect size was 0.20 for fixation instability values of the fellow eye and amblyopic eye under MV (total of 4 eye data [fellow eye and amblyopic eye in FEV and AEV conditions]), and 0.41 for fixation instability values under BV ([total of 2 eye data: fellow eye and amblyopic eye in BV]). To achieve a 95% power to detect significant differences in fixation instability between controls and amblyopia under BV, with a false-positive rate <5%, we would need to recruit 72 subjects, assuming an effect size of 0.41. For a more conservative estimate, we used the smaller effect size of 0.21. Consequently, to maintain 95% power with a false-positive rate <5%, we would have to recruit 132 subjects, ensuring robust detection of significant differences in fixation instability between controls and amblyopia, assuming the effect size is 0.21.

Data Recording and Feature Engineering

FEMs were preprocessed using a multistep algorithm to prepare them as input sequences for the DL model. The algorithm handled missing data by discarding segments with >1 second of consecutive missing samples and concatenating the remaining segments to maintain temporal continuity. For shorter gaps (≤1 second), linear interpolation was employed to fill in the missing data. In a subsequent step, the recordings were low-pass filtered at 50 Hz and downsampled by a factor of 5, resulting in 30-second sequences of 3000 data points sampled at 100 Hz. This downsampling improved the performance of the DL model while preserving the relevant information from the original recordings, including saccades, as their maximum frequency content is around 100 Hz.[63,64] Thus, although we collected eye positions for 45 seconds at 500 Hz, for analysis, we utilized only the first 30 seconds of denoised, deblinked, downsampled data during which the subject was fixating on the target.

The eye movement recordings from each subject consisted of 12 channels corresponding to horizontal and vertical eye positions of the right and left eyes recorded under the 3 viewing conditions: (1) BV with 4 channels; (2) MV with 8 channels (either the left or right eye viewing while the other eye is covered); and (3) BV + MV with 12 channels. The eye position traces acquired during visual fixation under MV and BV for each subject were transformed into rank 3 input tensor data structures with dimensions corresponding to the number of recording channels and the number of data points.[65]

Self-Attention–Based DL Model for FEM Data Analysis. A self-attention–based encoder model was implemented for this study as self-attention–based deep neural network have demonstrated significant improvement over other deep neural network architectures.[66] The self-attention mechanism focuses on links between sequences of data points and uses the relation between pairs of data points to rank features with higher importance.[66] This principle can be applied to eye movement recording, as the fixation of the viewing eye is likely to be dependent on its previous positions during a recording session. Therefore, a generalized self-attention mechanism was used to compute the weights of fixation stability values by comparing neighboring fixation stability in a sequence recording. Figure 3 illustrates the architectural components of the DL classifier used in this study. The DL model uses 8 independent heads (multihead attention mechanism) to concatenate the dense projections of the input fixation stability values from independent subspaces into a single sequence in output using global max pooling method.[67] This enables the classifier to use feature information from fixation stability values that may be correlated, for example recordings from a single subject under different viewing conditions, as compared to eye movement recorded from different subjects. Using the generalized self-attention architecture as reference, 2 additional dense projection layers were added to the output of the self-attention layer, followed by the addition of a residual connection layer and a layer normalization layer with the residual connection layer linked to the output of the layer normalization layer. Layer normalization is added to improve backpropagation gradient flow by normalizing the sequence of fixation stability values recorded in a batch independently.[67] The final classifier architecture uses multiple instances of encoder model that are stacked together with a dropout rate of 0.25 to generate the final classification result using a sigmoid function.

The model was implemented using the open-source TensorFlow Keras library, and it was trained with the Adam optimizer on an Apple M1 Pro GPU (16 GB RAM) for 50 epochs with a learning rate of 0.0001. We used k-fold cross-validation approach, with each fold serving as a test data set, while the remaining k-1 folds were used for training. The size and composition of the training and test datasets varied, corresponding to the different datasets representing various categories of patients; for example, a total of 72 subjects, with 32 patients with anisometropia and 40 control subjects were used (with k = 5, each fold had 14 to 15 subject) (detailed distribution of subjects is described in the Results section). This process was repeated k times, with each of the k folds used exactly once as the validation data (described in detail in the Results section). The hyperparameter values selected using the Keras tuning library and results are reported for all 3 viewing conditions (BV, MV, and BV + MV) in the next section.

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request. The underlying code for this study [and training/validation datasets] is not publicly available but may be made available to qualified researchers on reasonable request to the corresponding and senior author (F.G.).

Results

A total of 95 subjects with amblyopia (mean age, 13 [10.6] years; 51 women [53.6%]; 44 men [46.3%]) along with 40 controls (mean age, 14.90 [14.41] years; 22 women [57.5%]; 18 men [42.5%]) were included in the study. There were no differences in age (*t* test, $P = 0.39$) and sex (chi-square, $P = 0.68$) between controls vs. amblyopic subjects. Similarly, there was no difference in the distribution of race between controls (24 White [60%], 6 African American [16%]; 1 Hispanic [3%]; 5 Asian [13%]; 4 Multicultural [8%]) and amblyopia subjects (61 White [66%]; 12 African American [13%]; 10 Hispanic [11%]; 3 Asian [3%]; 6 Multi-cultural [7%]) (chi-square $P = 0.14$). The amblyopic cohort consisted of 32 anisometropic (7 treated, 6 mild, 16 moderate, and 3 severe), 34 mixed (7 treated, 2 mild, 16 moderate, and 9 severe), and 29 strabismic subjects (10 treated, 6 mild, 12 moderate, and 1 severe).

We report the results of the evaluation of classification accuracy across 3 datasets corresponding to subpopulations of patients selected based on the following: (1) types that is anisometropic, strabismic, and mixed amblyopia; (2) severity that is mild/treated and moderate/severe amblyopia; and (3) amblyopic subjects with or without nystagmus. The classification experiments were always binary, between 1 subpopulation of patients and controls. The results are the highest accuracy value reported by the transformer model. We also present the plots for the area under the receiver operating characteristic curve with y-axis corresponding to sensitivity (true-positive rate) and x-axis corresponding to 1-specificity (specificity representing the true-negative rate), together with the corresponding confusion matrix for viewing condition with the highest accuracy value[68] and the values of area under the precision–recall curve (AUPRC).

Classification of Subjects Based on the Type of Amblyopia

We have previously shown that FEM abnormalities are seen in amblyopic subjects regardless of the type of amblyopia. In anisometropic amblyopia, the abnormalities are more

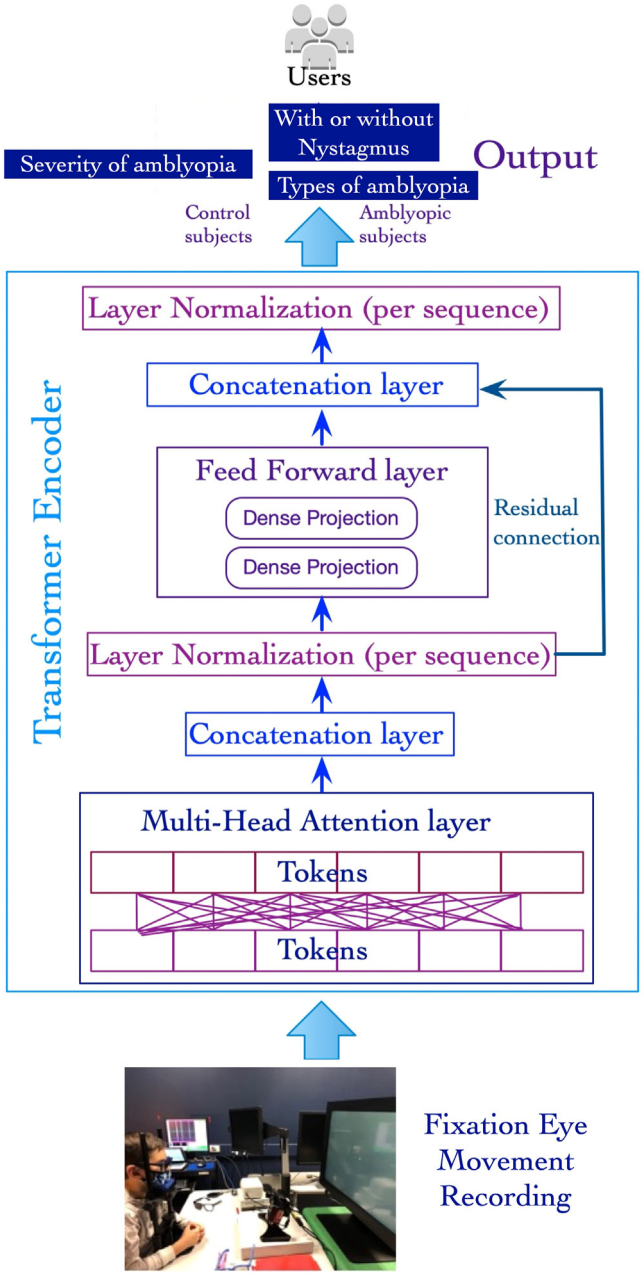


Figure 3. An overview of the deep neural network model architecture developed and used to classify patients with amblyopia from control subjects using fixation eye movement recordings.

pronounced in the amblyopic eye during BV and during monocular AEV compared to the fellow eye during monocular FEV condition. Conversely, in subjects with coexisting strabismus, the abnormalities are more pronounced in the nonviewing eye under MV and in the amblyopic eye during BV. Additionally, subjects with coexisting strabismus exhibit greater vergence instability across BV and MV. We created 3 datasets based on the type of amblyopia, that is, case I: anisometropic amblyopia subjects; case II: strabismus amblyopia subjects; case III: mixed amblyopic subjects with control subjects for each of the 3 cases. Table 1 lists the model performance under BV (4 channels), MV (8 channels), and BV + MV (12 channels) conditions for cases I, II, and III. The model achieved an average accuracy of 64% in case I (classification of anisometropic amblyopia), 74% in case II (classification of strabismic amblyopia), and 72% in case III (classification of mixed amblyopia) while utilizing horizontal and vertical eye position of the right and left eyes under BV (4 channels) with Figure 4 showing the corresponding area under the receiver operating characteristic plot and the confusion matrix for cases I, II, and III. The results show, as expected, that the model's discriminative abilities are greater in subjects

Table 1. Summary of the Performance of the Attentional Model for Classifying: (1) Patients with Anisometropic (Case I), Strabismus (Case II), and Mixed Types of Amblyopia (Case III); (2) Subpopulations of the Cohort with Mild or Treated Amblyopia (Case IV), Severe or Moderate Amblyopia (Case V); and (3) Amblyopic Patients with (Case VI) or without Nystagmus (Case VII) vs. Control Subjects across 3 Viewing Conditions

	Viewing Condition: BV (Number of Channels: 4)	Viewing Condition: MV (Number of Channels: 8)	Viewing Condition: BV + MV (Number of Channels: 12)
Dataset	Accuracy	Accuracy	Accuracy
Case I	0.64 (k = 4)	0.64 (k = 4)	0.62 (k = 6)
Case II	0.74 (k = 6)	0.73 (k = 5)	0.75 (k = 6)
Case III	0.72 (k = 6)	0.74 (k = 6)	0.73 (k = 6)
Case IV	0.73 (k = 6)	0.70 (k = 6)	0.65 (k = 6)
Case V	0.75 (k = 6)	0.72 (k = 6)	0.68 (k = 6)
Case VI	0.76 (k = 6)	0.76 (k = 5)	0.70 (k = 6)
Case VII	0.68 (k = 6)	0.64 (k = 5)	0.64 (k = 6)

BV = binocular viewing; MV = monocular viewing.

with coexisting strabismus than in those with anisometropic amblyopia. The values of AUPRC are reported in Table 2.

Classification of Subjects Based on the Severity of Amblyopic Eye Visual Acuity Deficit

Previous studies, including our own study, have shown that greater visual acuity deficit of the amblyopic eye is associated with greater FEM abnormalities in the amblyopic eye during monocular AEV conditions.[30,37] To evaluate the capability of the model to classify subjects according to their severity, we created 2 datasets, that is, case IV consisting of subjects with either treated or mild amblyopia (n = 37) and case V consisting of subjects with moderate or severe amblyopia (n = 57), with control subjects for each of the 2 cases. Table 1 lists the model performance under BV, MV, and BV + MV conditions for cases IV and V. Figure 5 shows the area under the receiver operating characteristic plot and the confusion

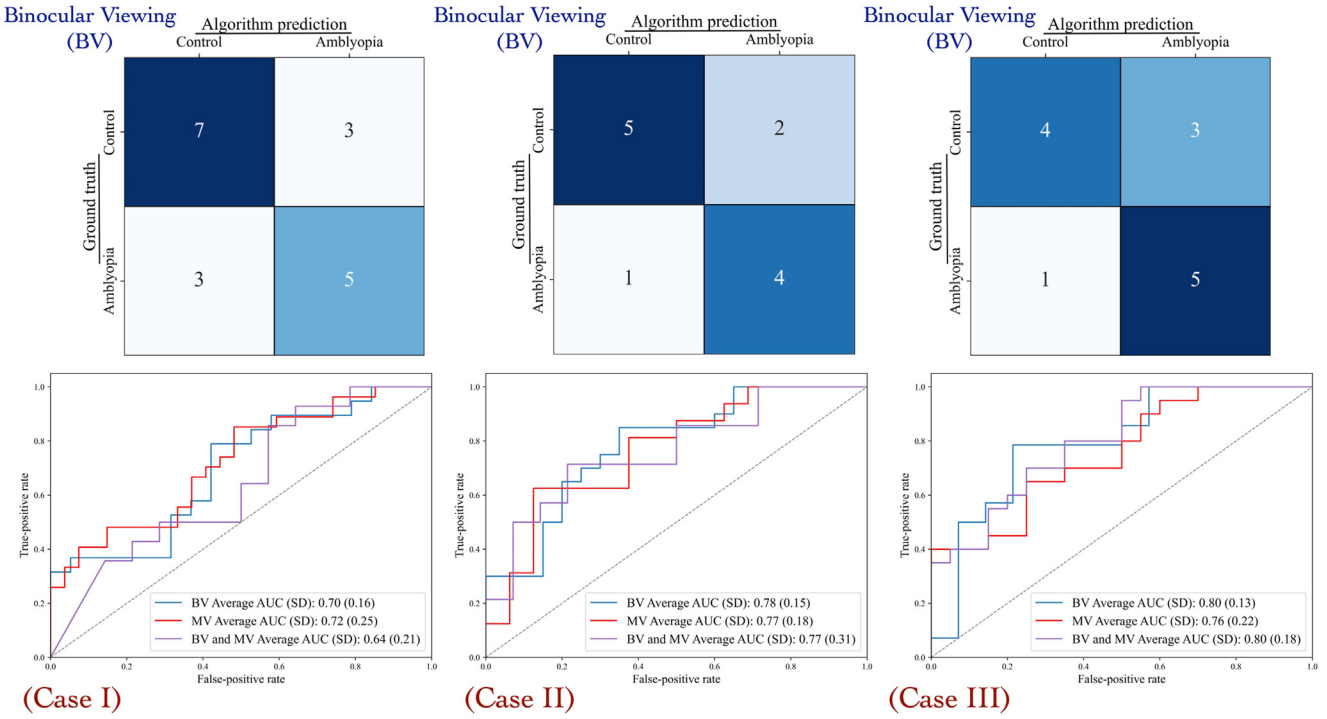


Figure 4. The classification of subjects with **(A)** anisometropia (case I), **(B)** strabismus (case II), and mixed (case III). The confusion matrix represents the ground truth and the values predicted by the deep learning algorithm. For each case, the AUC traces represent the performance of the neural network across all 3 viewing conditions. AUC = area under curve; BV = binocular viewing; MV = monocular viewing; SD = standard deviation.

Table 2. Area Under the Precision–Recall Curve of the Performance of the Attentional Model for Classifying: (1) Patients with Anisometropic (Case I), Strabismus (Case II), and Mixed Types of Amblyopia (Case III); (2) Subpopulations of the Cohort with Mild or Treated Amblyopia (Case IV), Severe or Moderate Amblyopia (Case V); and (3) Amblyopic Patients with (Case VI) or without Nystagmus (Case VII) vs. Control Subjects across 3 Viewing Conditions

Model	Viewing Condition: BV + MV (Number of Channels: 12)	Viewing Condition: MV (Number of Channels: 8)	Viewing Condition: BV (Number of Channels: 4)
Case I	0.64 (0.11)	0.69 (0.10)	0.72 (0.08)
Case II	0.78 (0.12)	0.77 (0.14)	0.81 (0.16)
Case III	0.79 (0.14)	0.75 (0.12)	0.82 (0.15)
Case IV	0.66 (10)	0.73 (0.13)	0.76 (0.18)
Case V	0.74 (0.08)	0.77 (0.13)	0.79 (0.16)
Case VI	0.78 (0.13)	0.83 (0.18)	0.81 (0.18)
Case VII	0.74 (0.24)	0.72 (0.22)	0.76 (0.09)

BV = binocular viewing; MV = monocular viewing.
Values in parentheses are standard deviation.

matrix of the model, which achieves an accuracy of 73% and 75%, respectively, in classifying subjects with mild/treated and those with moderate/severe amblyopia from controls under BV. The findings suggest that the model's discriminative capabilities are similar when comparing subjects with moderate/severe amblyopia to controls, as well as when comparing those with treated/mild amblyopia to controls. This similarity may be attributed to the persistent FEM abnormalities observed in individuals with previously treated amblyopia, which could explain these results.[44,45] The corresponding values of AUPRC are included in Table 2.

Classification of Subjects Involving Amblyopic Subjects with and without Nystagmus

We have previously shown that FEM abnormalities arise in amblyopia due to alterations of physiologic FEMs with increased amplitude of fast FEMs and increased velocity of slow FEMs or nystagmus, which are associated with marked increase in frequency and amplitude of fast and slow FEMs compared to controls. Thus, in the final phase of this study, we created 2 datasets corresponding to amblyopic subjects with nystagmus case VI (n = 53) and case VII: amblyopic subjects without nystagmus (n = 37), each with control subjects. Table 1 also lists the performance of the model under MV and BV + MV viewing conditions for cases VI and VII. Figure 6 shows the corresponding area under curve plot and the confusion matrix of the model, which achieves an accuracy of 76% and 68%, respectively, in classifying subjects with nystagmus and those without nystagmus under BV. Thus, the results show, as expected, that the model's discriminative abilities are greater in subjects with coexisting nystagmus than in those without nystagmus. The values of AUPRC for cases VI and VII are included in Table 2.

Discussion

This study describes a multihead attention-based DL algorithm designed to analyze horizontal and vertical eye positions during visual fixation tasks using infrared video-oculography from control and amblyopic subjects. The algorithm successfully classified amblyopic subjects based on the type and severity of amblyopia. We also found that the model had better discriminative capabilities in distinguishing subjects with nystagmus than those without nystagmus.

Recent advancements in DL have shown promise in detecting diabetic retinopathy and retinopathy of prematurity and in automating the diagnosis and prescreening of adult ophthalmic conditions. In pediatric ophthalmology, where specialists are scarce,[69,70] machine learning models have been developed to detect strabismus using photographs,[71–73] cover test videos,[74] and prism cover test measurements[75] from pediatric eye clinics. Studies employing convolutional neural networks pretrained on datasets like ImageNet have analyzed primary gaze photographs,[73] and eye-tracking data,[76] demonstrating high specificity and varying sensitivity in identifying strabismus. Additional research utilizing retinal birefringence scanning and convolutional neural network models has identified strabismus in smaller cohorts.[77] Machine learning has also shown promising results in patients with congenital and acquired nystagmus.[78–80] One study used electro-oculography to record eye movements in patients with congenital motor nystagmus, investigating relationships among different nystagmus parameters to predict visual acuity. Another study applied a decision tree model to detect amblyogenic risk factors in young children using video frames from an instrument-based screening device, achieving 77% accuracy, particularly in identifying strabismic subjects and those with significant refractive errors.[81] Overall, these studies highlight the potential of machine learning to enhance the diagnosis and management of pediatric ophthalmic conditions, including strabismus, nystagmus, and amblyopia risk factors.

Previous research has highlighted increased instability of the amblyopic eye relative to the fellow eye in individuals with amblyopia compared to controls. These abnormalities are seen in subjects with coexisting strabismus and are also observed in those with anisometropic amblyopia. On the other hand, FEM abnormalities such as increased vergence instability and increased fixation instability, are more

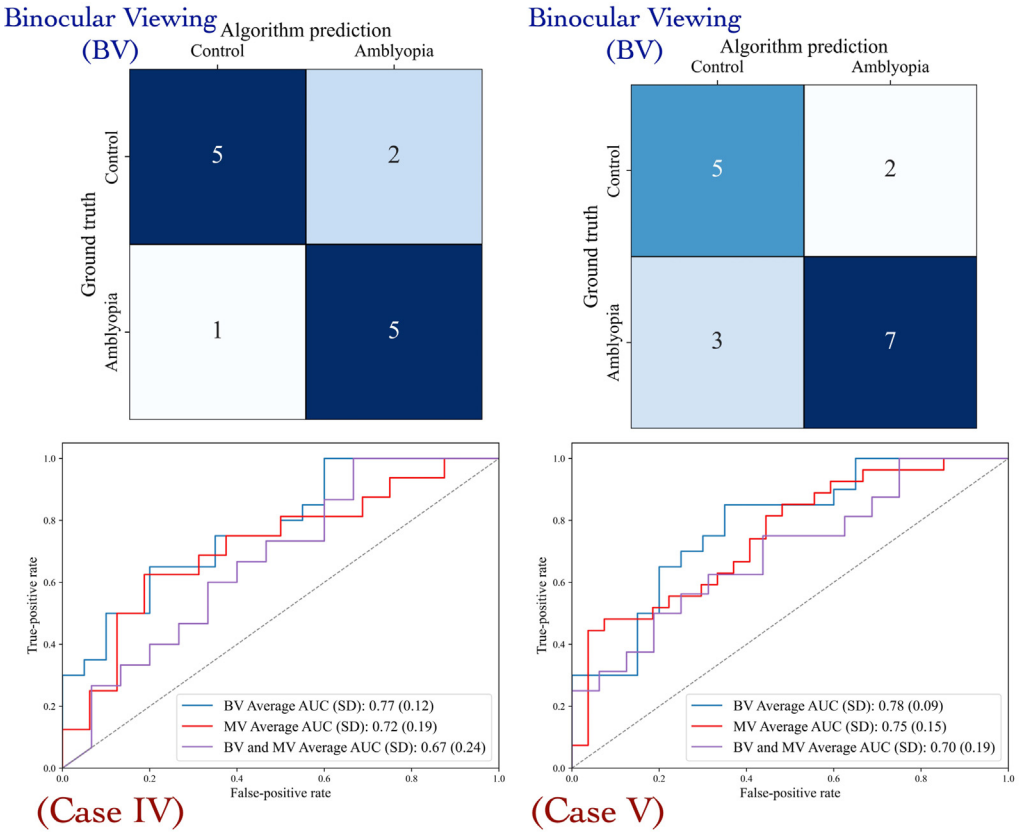


Figure 5. Classification results for subjects with mild or treated amblyopia (case IV) and patients with moderate or severe amblyopia (case V). The confusion matrix represents the ground truth and the values predicted by the deep learning algorithm. The AUC traces represent the performance of the neural network across all 3 viewing conditions. AUC = area under curve; BV = binocular viewing; MV = monocular viewing; SD = standard deviation.

pronounced in the deviated eye of individuals with strabismic or mixed amblyopia under MV and BV compared to anisometropic amblyopia subjects. Further, strabismic/mixed amblyopia are more likely to have nystagmus,[45] contributing to superior model performance in detecting these subtypes within our study. Our DL model performance was comparable in subjects with mild or previously treated as well as moderate or severe amblyopia. This could be attributed to significant FEM abnormalities in mild/treated amblyopic subjects such as increased fixation instability, increased amplitude of fast FEMs, and increased drift velocities observed in both groups of patients under BV and MV compared to controls.[30,37] Our study found that DL models achieve comparable performance using eye position data from BV alone. Our recent work has demonstrated that interocular fixation instability observed in amblyopia are not affected by age.[30,82] Thus, within our current study, we incorporated horizontal and vertical eye positions acquired during fixation of the right and left eye under binocular, monocular right eye, and monocular left eye viewing conditions. This approach allowed the model to learn from patterns of interocular fixation instability observed in amblyopia compared to controls. Thus, within our current study, we did not find any trend in accuracy with respect to age across all the 7 cases described above. Thus, the current study highlights the ability of the trained DL model to effectively distinguish amblyopic patients from control subjects with a high degree of accuracy, which may be translated into a clinical informatics tool for clinical use.

In summary, DL models that utilize FEM data can help in development of automated clinical informatics tools in identifying subjects with amblyopia. Strabismic/mixed amblyopia shows more pronounced FEM abnormalities, leading to better model performance compared to anisometropic amblyopia. Using BV data alone are sufficient for accurate prediction, simplifying the testing process and reducing time, especially beneficial in clinical pediatric assessments.

Our study has several limitations. First, the potential for selection bias exists, as the participants were children with known amblyopia who could perform eye movement recordings, which may not represent the general population diversity. Future studies in general population settings, where amblyopia prevalence is lower, could provide a more representative sample. These studies would provide insights into how eye movement abnormalities can identify amblyopia in a broader context, beyond clinically enriched populations. The eye tracker utilized in the present study is a high-grade, costly research instrument designed for laboratory-based investigations. Future research should

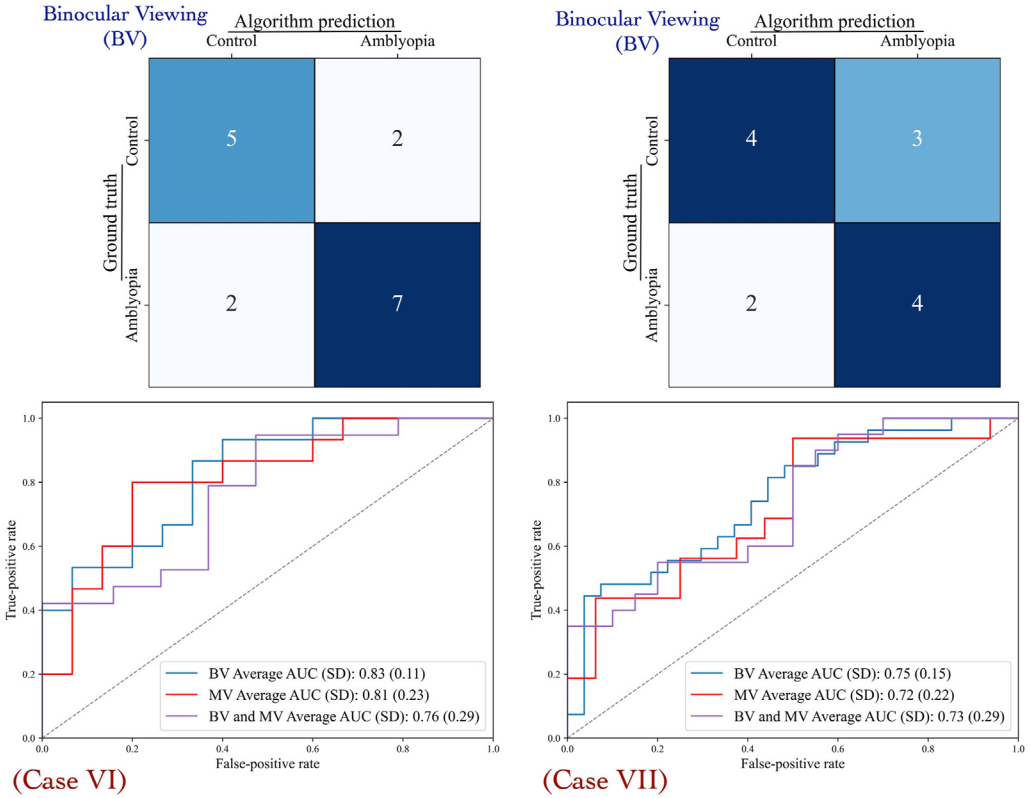


Figure 6. The classification of amblyopic subjects **(A)** with nystagmus (case VI) and **(B)** without nystagmus (case VII). The confusion matrix represents the ground truth and the values predicted by the deep learning algorithm. The AUC traces represent the performance of the neural network across all 3 viewing conditions. AUC = area under curve; BV = binocular viewing; MV = monocular viewing; SD = standard deviation.

explore the potential of instrument-based screening devices in conjunction with eye movement data obtained from more portable and affordable eye trackers. This combined approach has the potential to enhance classification accuracy as compared to current methods that often exhibit low accuracy. This approach would be useful for screenings in schools, pediatrician offices, and pediatric eye care practices, where young children may struggle with traditional visual acuity tests. Also, DL networks need uncontaminated, high-quality eye position data free from blinks and noise, with subjects fixing on the target. The methods and code used for data analysis in this study integrate both automated and user-interactive interfaces to enable acquiring high-quality fixation eye position data. Our future work will focus on developing a comprehensive pipeline that incorporates customized calibration parameters, such as screen size, viewing distance, and sampling frequencies, as well as data preprocessing and visualization techniques. This pipeline will also integrate the automated Engbert and unsupervised clustering methods for FEM analysis, alongside new DL algorithms. Such advancements will streamline the implementation of eye movement analysis following the acquisition of recordings from various eye trackers, thereby enhancing the practical application of these techniques in clinical settings. Previous studies evaluating FEMs in amblyopia have used shorter duration of fixation ranging from 6 to 45 seconds and have reported FEM abnormalities[34,35,40,61] compared to controls. Our trials for the current study were structured to last 45 seconds to ensure adequate collection of high-quality data. However, for analysis, we utilized only the first 30 seconds of denoised, deblinked data, during which the subject was fixating on the target. In our future studies, we will evaluate whether fixation eye position data acquired for shorter periods of time than used in the current study may be adequate for identifying amblyopia. Also, in our current study, the data were downsampled from 500 Hz to 100 Hz. The model's strong performance with lower-frequency data demonstrates that less expensive, lower-sampling eye trackers could be effective in outpatient clinical settings, potentially yielding comparable results. If realized, this scenario holds promise for expanding eye movement analysis applications in clinical practice, improving diagnostic capabilities for conditions such as amblyopia, strabismus, and nystagmus. Another limitation of this study is the exclusion of deprivation amblyopia, which is less common than anisometropic and strabismic amblyopia. Including it could introduce variability due to its distinct causes and presentation, potentially skewing results given the limited sample size. This exclusion helps maintain a homogeneous dataset, enhancing the accuracy and generalizability of the artificial intelligence algorithms for identifying the more common forms of amblyopia. Finally, the study's single-site design may limit the generalizability of the findings,

highlighting the need for multicenter trials to validate results across diverse populations and settings.

This study is a pioneering effort to explore using eye movement abnormalities for identifying amblyopia in a large cohort. We demonstrated that eye positions captured during visual fixation using infrared video oculography characterize amblyopic subjects and that transformer models can extract features from eye-tracking data for classification. Our results provide a foundation for developing assistive and potentially automated tools for the identification of amblyopia. With recent hardware and software advancements in compact, accurate eye-tracking technology, future research should focus on large FEM datasets, advanced algorithms, and assessing the generalizability, clinical utility, and cost-effectiveness of DL models in various settings.

Footnotes and Disclosures

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Obtained funding: Ghasia

Overall responsibility: Upadhyaya, Cakir, Ramat, Shaikh, Sahoo, Ghasia

Abbreviations and Acronyms:
AEV = amblyopic eye viewing; **AUPRC** = area under the precision−recall curve; **BV** = binocular viewing; **D** = diopters; **DL** = deep learning; **FEM** = fixation eye movement; **FEV** = fellow eye viewing; **FMN** = fusion maldevelopment nystagmus; **MV** = monocular viewin.

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References

1. Friedman DS, Repka MX, Katz J, et al. Prevalence of amblyopia and strabismus in white and African American children aged 6 through 71 months the Baltimore pediatric eye disease study. *Ophthalmology*. 2009;116:2128−2134.e1-2.
2. Hu B, Liu Z, Zhao J, et al. The global prevalence of amblyopia in children: a systematic review and meta-analysis. *Front Pediatr*. 2022;10:819998.
3. Mostafaie A, Ghojazadeh M, Hosseinifard H, et al. A systematic review of Amblyopia prevalence among the children of the world. *Rom J Ophthalmol*. 2020;64(4):342−355.
4. Fu Z, Hong H, Su Z, et al. Global prevalence of amblyopia and disease burden projections through 2040: a systematic review and meta-analysis. *Br J Ophthalmol*. 2020;104:1164−1170.
5. Hashemi H, Pakzad RM, Yekta A, et al. Global and regional estimates of prevalence of amblyopia: a systematic review and meta-analysis. *Strabismus*. 2018;26:168−183.
6. Wong AM. New concepts concerning the neural mechanisms of amblyopia and their clinical implications. *Can J Ophthalmol*. 2012;47:399−409.

7. Grant S, Moseley MJ. Amblyopia and real-world visuomotor tasks. *Strabismus*. 2011;19:119–128.
8. Kanonidou E, Proudlock FA, Gottlob I. Reading strategies in mild to moderate strabismic amblyopia: an eye movement investigation. *Invest Ophthalmol Vis Sci*. 2010;51:3502–3508.
9. Kelly KR, Jost RM, De La Cruz A, et al. Slow reading in children with anisometropic amblyopia is associated with fixation instability and increased saccades. *J AAPOS*. 2017;21:447–451.e1.
10. Niechwiej-Szwedo E, Goltz HC, Chandrakumar M, et al. Effects of anisometropic amblyopia on visuomotor behavior, I: saccadic eye movements. *Invest Ophthalmol Vis Sci*. 2010;51:6348–6354.
11. Chen D, Otero-Millan J, Kumar P, et al. Visual search in amblyopia: abnormal fixational eye movements and suboptimal sampling strategies. *Invest Ophthalmol Vis Sci*. 2018;59:4506–4517.
12. Bhutada ISP, Jacobs J, Murray J, et al. Reading difficulties in amblyopia: consequence of visual sensory and oculomotor dysfunction. *J Neurol Sci*. 2022;442:120438.
13. Grant S, Suttle C, Melmoth DR, et al. Age- and stereovision-dependent eye-hand coordination deficits in children with amblyopia and abnormal binocularity. *Invest Ophthalmol Vis Sci*. 2014;55:5687–57015.
14. Kelly KR, Morale SE, Beauchamp CL, et al. Factors associated with impaired motor skills in strabismic and anisometropic children. *Invest Ophthalmol Vis Sci*. 2020;61:43.
15. Niechwiej-Szwedo E, Colpa L, Wong AMF. Visuomotor behaviour in amblyopia: deficits and compensatory adaptations. *Neural Plast*. 2019;2019:6817839.
16. Carlton J, Kaltenthaler E. Amblyopia and quality of life: a systematic review. *Eye (Lond)*. 2011;25:403–413.
17. Assaf AA. The sensitive period: transfer of fixation after occlusion for strabismic amblyopia. *Br J Ophthalmol*. 1982;66:64–70.
18. Williams CN, Northstone K, Harrad RA, et al. Amblyopia treatment outcomes after screening before or at age 3 years: follow up from randomized trial. *BMJ*. 2002;324:1549–1551.
19. Woodruff G, Hiscox F, Thompson JR, Smith LK. The presentation of children with amblyopia. *Eye (Lond)*. 1994;8(Pt 6):623–626.
20. Murali K, Vidhya C, Murthy SR, Mallapa S. Cost-Effectiveness of photoscreeners in screening at-risk amblyopia in Indian children. *Indian J Public Health*. 2022;66:171–175.
21. Carlton J, Karnon J, Czoski-Murray C, et al. The clinical effectiveness and cost-effectiveness of screening programmes for amblyopia and strabismus in children up to the age of 4-5 years: a systematic review and economic evaluation. *Health Technol Assess*. 2008;12:iii. xi-194.
22. Bregman J, Donahue SP. Validation of photoscreening technology in the general pediatrics office: a prospective study. *J AAPOS*. 2016;20:153–158.
23. Nishimura M, Wong A, Cohen A, et al. Choosing appropriate tools and referral criteria for vision screening of children aged 4-5 years in Canada: a quantitative analysis. *BMJ Open*. 2019;9:e032138.
24. Langeslag-Smith MA, Vandal AC, Briane V, et al. Preschool children's vision screening in New Zealand: a retrospective evaluation of referral accuracy. *BMJ Open*. 2015;5:e009207.
25. Schmucker C, Grosselfinger R, Riemsma R, et al. Effectiveness of screening preschool children for amblyopia: a systematic review. *BMC Ophthalmol*. 2009;9:3.
26. Williams C, Harrad RA, Harvey I, et al. Screening for amblyopia in preschool children: results of a population-based, randomised controlled trial. ALSPAC Study Team. Avon Longitudinal Study of Pregnancy and Childhood. *Ophthalmic Epidemiol*. 2001;8:279–295.
27. Preslan MW, Novak A. Baltimore vision screening project. *Ophthalmology*. 1996;103:105–109.
28. Dobson V, Teller DY. Visual acuity in human infants: a review and comparison of behavioral and electrophysiological studies. *Vis Res*. 1978;18:1469–1483.
29. Quinn GE, Berlin JA, James M. The Teller acuity card procedure. Three testers in a clinical setting. *Ophthalmology*. 1993;100:488–494.
30. Ghasia F, Tychsen L. Inter-ocular fixation instability of amblyopia: relationship to visual acuity, strabismus, nystagmus, stereopsis, vergence and age. *Am J Ophthalmol*. 2024;267:230–248.
31. Martinez-Conde S, Macknik SL, Troncoso XG, Dyar TA. Microsaccades counteract visual fading during fixation. *Neuron*. 2006;49:297–305.
32. McCamy MB, Otero-Millan J, Leigh RJ, et al. Simultaneous recordings of human microsaccades and drifts with a contemporary video eye tracker and the search coil technique. *PLoS One*. 2015;10:e0128428.
33. Chung ST, Kumar G, Li RW, Levi DM. Characteristics of fixational eye movements in amblyopia: limitations on fixation stability and acuity? *Vis Res*. 2015;114:87–99.
34. Subramanian V, Jost RM, Birch EE. A quantitative study of fixation stability in amblyopia. *Invest Ophthalmol Vis Sci*. 2013;54:1998–2003.
35. Gonzalez EG, Wong AM, Niechwiej-Szwedo E, et al. Eye position stability in amblyopia and in normal binocular vision. *Invest Ophthalmol Vis Sci*. 2012;53:5386–5394.
36. Shaikh AG, Otero-Millan J, Kumar P, Ghasia FF. Abnormal fixational eye movements in amblyopia. *PLoS One*. 2016;11:e0149953.
37. Kang SL, Beylergil SB, Otero-Millan J, et al. Fixational eye movement waveforms in amblyopia: characteristics of fast and slow eye movements. *J Eye Mov Res*. 2019;12:9.
38. Ciuffreda KJ, Kenyon RV, Stark L. Fixational eye movements in amblyopia and strabismus. *J Am Optom Assoc*. 1979;50:1251–1258.
39. Scaramuzzi M, Murray J, Otero-Millan J, et al. Fixation instability in amblyopia: oculomotor disease biomarkers predictive of treatment effectiveness. *Prog Brain Res*. 2019;249:235–248.
40. Shi XF, Xu LM, Li Y, et al. Fixational saccadic eye movements are altered in anisometropic amblyopia. *Restor Neurol Neurosci*. 2012;30:445–462.
41. Economides JR, Adams DL, Horton JC. Variability of ocular deviation in strabismus. *JAMA Ophthalmol*. 2016;134:63–69.
42. Scaramuzzi M, Murray J, Nucci P, et al. Fixational eye movements abnormalities and rate of visual acuity and stereoacuity improvement with part time patching. *Sci Rep*. 2021;11:1217.
43. Murray J, Garg K, Ghasia F. Monocular and binocular visual function deficits in amblyopic patients with and without fusion maldevelopment nystagmus. *Eye Brain*. 2021;13:99–109.
44. Dulaney CS, Murray J, Ghasia F. Contrast sensitivity, optotype acuity and fixation eye movement abnormalities in amblyopia under binocular viewing. *J Neurol Sci*. 2023;451:120721.
45. Cakir GB, Murray J, Dulaney C, Ghasia F. Multifaceted interactions of stereoacuity, inter-ocular suppression, and fixation eye movement abnormalities in amblyopia and strabismus. *Invest Ophthalmol Vis Sci*. 2024;65:19.
46. Murray J, Gupta P, Dulaney C, et al. Effect of viewing conditions on fixation eye movements and eye alignment in amblyopia. *Invest Ophthalmol Vis Sci*. 2022;63:33.

47. Gulshan V, Peng L, Coram M, et al. Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. *JAMA*. 2016;316:2402−2410.
48. Zemblys R, Niehorster DC, Komogortsev O, Holmqvist K. Using machine learning to detect events in eye-tracking data. *Behav Res Methods*. 2018;50:160−181.
49. Friedman L, Rigas I, Abdulin E, Komogortsev OV. A novel evaluation of two related and two independent algorithms for eye movement classification during reading. *Behav Res Methods*. 2018;50:1374−1397.
50. Dell'Osso LF, Daroff RB. Congenital nystagmus waveforms and foveation strategy. *Doc Ophthalmol*. 1975;39:155−182.
51. Manh VM, Holmes JM, Lazar EL, et al. A randomized trial of a binocular iPad game versus part-time patching in children aged 13 to 16 Years with amblyopia. *Am J Ophthalmol*. 2018;186:104−115.
52. Chen AM, Cotter SA. The amblyopia treatment studies: implications for clinical practice. *Adv Ophthalmol Optom*. 2016;1:287−305.
53. Cotter SA, Pediatric Eye Disease Investigator G, Edwards AR, et al. Treatment of anisometropic amblyopia in children with refractive correction. *Ophthalmology*. 2006;113:895−903.
54. Wallace DK, Pediatric Eye Disease Investigator G, Edwards AR, et al. A randomized trial to evaluate 2 hours of daily patching for strabismic and anisometropic amblyopia in children. *Ophthalmology*. 2006;113:904−912.
55. Tang S, Skelly P, Otero-Millan J, et al. Effects of visual blur on microsaccades during visual exploration. *J Eye Mov Res*. 2019;12:10.
56. Simon JW, Siegfried JB, Mills MD, et al. A new visual evoked potential system for vision screening in infants and young children. *J Aapos*. 2004;8:549−554.
57. Engbert R, Kliegl R. Microsaccades uncover the orientation of covert attention. *Vis Res*. 2003;43:1035−1045.
58. Engbert R, Kliegl R. Microsaccades keep the eyes' balance during fixation. *Psychol Sci*. 2004;15:431−436.
59. Laubrock J, Engbert R, Kliegl R. Microsaccade dynamics during covert attention. *Vis Res*. 2005;45:721−730.
60. Steinman RM, Cushman WB, Martins AJ. The precision of gaze. A review. *Hum Neurobiol*. 1982;1(2):97−109.
61. Ghasia F, Wang J. Amblyopia and fixation eye movements. *J Neurol Sci*. 2022;441:120373.
62. Upadhyaya S, Pullela M, Ramachandran S, et al. Fixational saccades and their relation to fixation instability in strabismic monkeys. *Invest Ophthalmol Vis Sci*. 2017;58:5743−5753.
63. Inchingolo P, Spanio M. On the identification and analysis of saccadic eye movements–a quantitative study of the processing procedures. *IEEE Trans Biomed Eng*. 1985;32:683−695.
64. Bahill AT, Brockenbrough A, Troost BT. Variability and development of a normative data base for saccadic eye movements. *Invest Ophthalmol Vis Sci*. 1981;21(1 Pt 1):116−125.
65. Kossaifi J, Panagakis Y, Anandkumar A, Pantic M. Tensorly: tensor learning in python. *J Machine Learn Res*. 2019;20:1−6.
66. Vaswani A, Shazeer N, Parmar N, et al. Attention is all you need. *Adv Neural Inf Process Syst*. 2017;30.
67. Chollet F. *Deep learning with Python*. Boston: Simon and Schuster; 2021.
68. Zou KH, O'Malley AJ, Mauri L. Receiver-operating characteristic analysis for evaluating diagnostic tests and predictive models. *Circulation*. 2007;115:654−657.
69. Reid JE, Eaton E. Artificial intelligence for pediatric ophthalmology. *Curr Opin Ophthalmol*. 2019;30:337−346.
70. Estes R, Estes D, West C, et al. The American association for pediatric ophthalmology and strabismus workforce distribution project. *J AAPOS*. 2007;11:325−329.
71. Yang HK, Seo JM, Hwang JM, Kim KG. Automated analysis of binocular alignment using an infrared camera and selective wavelength filter. *Invest Ophthalmol Vis Sci*. 2013;54:2733−2737.
72. Sousa de Almeida JD, Silva AC, Teixeira JA, et al. Computer-aided methodology for syndromic strabismus diagnosis. *J Digit Imaging*. 2015;28:462−473.
73. Zheng C, Yao Q, Lu J, et al. Detection of referable horizontal strabismus in children's primary gaze photographs using deep learning. *Transl Vis Sci Technol*. 2021;10:33.
74. Valente TLA, de Almeida JDS, Silva AC, et al. Automatic diagnosis of strabismus in digital videos through cover test. *Comput Methods Programs Biomed*. 2017;140:295−305.
75. Chandna A, Fisher AC, Cunningham I, et al. Pattern recognition of vertical strabismus using an artificial neural network (StrabNet). *Strabismus*. 2009;17:131−138.
76. Chen G, Xu R, Yang Z. Deep ranking structural support vector machine for image tagging. *Pattern Recognit Lett*. 2018;105:30−38.
77. Gramatikov B. Detecting fixation on a target using time-frequency distributions of a retinal birefringence scanning signal. *Biomed Eng Online*. 2013;12:41.
78. D'Addio G, Ricciardi C, Improta G, et al. *Feasibility of machine learning in predicting features related to congenital nystagmus. IFMBE Proceedings book series*. 2020.
79. Lim EC, Park JH, Jeon HJ, et al. Developing a diagnostic decision support system for benign paroxysmal positional vertigo using a deep-learning model. *J Clin Med*. 2019;8:633.
80. Martti JHA, Joutsijoki H, Hirvonen TP. The classification of valid and invalid beats of three-dimensional nystagmus eye movement signals using machine learning methods. *Adv Arti cial Neural Syst*. 2013.
81. Van Eenwyk J, Agah A, Giangiacomo J, Cibis G. Artificial intelligence techniques for automatic screening of amblyogenic factors. *Trans Am Ophthalmol Soc*. 2008;106:64−73; discussion 73-4.
82. Yue SC, Cakir GB, Shaikh A, Ghasia FF. Assessing interocular fixational eye movements throughout the lifespan. *Exp Brain Res*. 2024;242:2749−2763.